# Trend Analysis of Cropping Patterns in Mancherial District (2016–2019)

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Course: Data Analytics and Visualization | Instructor: Dr. Amit Swamy

## Abstract

This study examines agricultural trends and cropping pattern shifts in Mancherial District from 2016–2019 using government crop area data for Kharif and Rabi seasons. Using Power BI for visualization, key patterns, correlations, and seasonal variations were analyzed to understand land-use dynamics and support future planning.

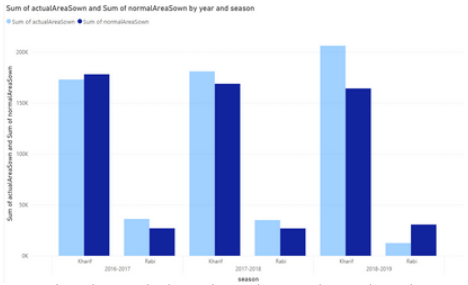



Fig 1: The Graph above shows the Actual area planted vs normal area throught 3 years

## Introduction

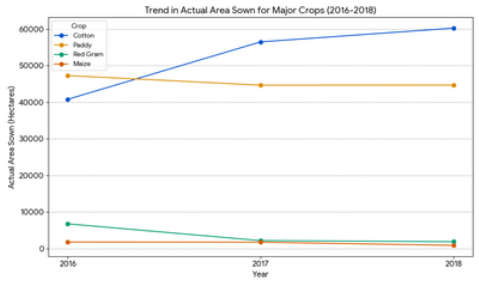
Agricultural planning is fundamentally challenged by market price volatility and climate variability, this requires a deep understanding of dynamic cropping pattern shifts and farmers underlying flexibility in resource allocation. This study analyzes crop area data from the Mancherial district spanning 2016 to 2019 to investigate the evolution of the local agricultural landscape.

There are 3 main Objectives for this study:

- To conduct a Trend Analysis of major crop area and identify the dominant crop in the region.
- To indentify the short-term Volatility in area sown for key crops.
- To derive Policy implications and Forecasting insights based on observed shifts

By defining the interplay between stable crops, high-value crops, and volatile 'swing' crops, this research provides a crucial evidence base for informed policy interventions

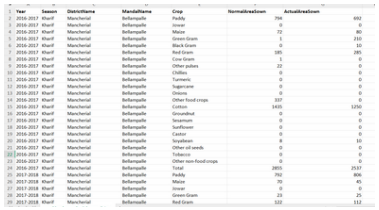
Fig 2: Major crops sown in Mancherial District 2016-18




## Materials

The dataset used in this study includes the area used for sowing different crops and the long time average compared to the current area sown in that year.

The tools used in this study are: Power Bi for visualising the data , Microsoft Excel for standardizing the data as the Mandal names in each of the datasets was either wrong or used a different variation of the spelling, this change decreased the 33 Mandals in the original dataset to only 18 Mandals.



| | Year | Season | DistrictName | MandalName | Crop | NormalAreaSown | ActualAreaSown |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | 2016-2017 | Kharif | Mancherial | Bellampalle | Paddy | 794 | 692 |
| 3 | 2016-2017 | Kharif | Mancherial | Bellampalle | Jowar | 0 | 0 |
| 4 | 2016-2017 | Kharif | Mancherial | Bellampalle | Maize | 72 | 80 |
| 5 | 2016-2017 | Kharif | Mancherial | Bellampalle | Green Gram | 1 | 210 |
| 6 | 2016-2017 | Kharif | Mancherial | Bellampalle | Black Gram | 0 | 10 |
| 7 | 2016-2017 | Kharif | Mancherial | Bellampalle | Red Gram | 185 | 295 |
| 8 | 2016-2017 | Kharif | Mancherial | Bellampalle | Cow Gram | 1 | 0 |
| 9 | 2016-2017 | Kharif | Mancherial | Bellampalle | Other pulses | 22 | 0 |
| 10 | 2016-2017 | Kharif | Mancherial | Bellampalle | Chillies | 0 | 0 |
| 11 | 2016-2017 | Kharif | Mancherial | Bellampalle | Turmeric | 0 | 0 |
| 12 | 2016-2017 | Kharif | Mancherial | Bellampalle | Sugarcane | 0 | 0 |
| 13 | 2016-2017 | Kharif | Mancherial | Bellampalle | Onions | 0 | 0 |
| 14 | 2016-2017 | Kharif | Mancherial | Bellampalle | Other food crops | 337 | 0 |
| 15 | 2016-2017 | Kharif | Mancherial | Bellampalle | Cotton | 1435 | 1250 |
| 16 | 2016-2017 | Kharif | Mancherial | Bellampalle | Groundnut | 0 | 0 |
| 17 | 2016-2017 | Kharif | Mancherial | Bellampalle | Sesamum | 0 | 0 |
| 18 | 2016-2017 | Kharif | Mancherial | Bellampalle | Sunflower | 0 | 0 |
| 19 | 2016-2017 | Kharif | Mancherial | Bellampalle | Castor | 0 | 0 |
| 20 | 2016-2017 | Kharif | Mancherial | Bellampalle | Soyabean | 8 | 10 |
| 21 | 2016-2017 | Kharif | Mancherial | Bellampalle | Other oil seeds | 0 | 0 |
| 22 | 2016-2017 | Kharif | Mancherial | Bellampalle | Tobacco | 0 | 0 |
| 23 | 2016-2017 | Kharif | Mancherial | Bellampalle | Other non-food crops | 0 | 0 |
| 24 | 2016-2017 | Kharif | Mancherial | Bellampalle | Total | 2855 | 2537 |
| 25 | 2017-2018 | Kharif | Mancherial | Bellampalle | Paddy | 792 | 806 |
| 26 | 2017-2018 | Kharif | Mancherial | Bellampalle | Maize | 70 | 45 |
| 27 | 2017-2018 | Kharif | Mancherial | Bellampalle | Jowar | 0 | 0 |
| 28 | 2017-2018 | Kharif | Mancherial | Bellampalle | Green Gram | 23 | 25 |
| 29 | 2017-2018 | Kharif | Mancherial | Bellampalle | Red Gram | 122 | 112 |

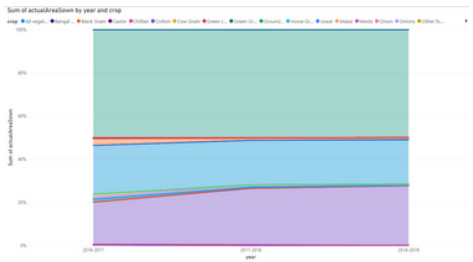
Fig 3: Cleaned and standardized dataset of all 3 years combined into one

## Methodology

The analysis was divided into Five topics for clarity and focus:

1. **Trend Analysis**: Values of Actual area sown for the top 4 crops were plotted over time to identify linear growth or decline. (Fig 1)
2. **Cropping Pattern Shifts**: The annual area of each crop was calculated as a percentage share of the Total Area Sown. This relative metric was visualized using a stacked area chart to clearly illustrate the trade-offs and proportional shifts between crops.
3. **Correlation and Volatility**: Correlation was observed by comparing the year-over-year changes in the percentage share of dominant crops. Volatility (risk) was quantified using the Coefficient of Variation ((CV)= {Standard Deviation} / {Mean}) on the annual area sown
4. **Policy Implications and Forecasting**: Trends were extrapolated, and volatility results were used to assign risk profiles to each crop
5. **Flexibility :** Flexibility was measured by the responsiveness and magnitude of change in the 'swing' crop

Fig 4: Annual area of each crop calculated as a percentage of the total




## Results

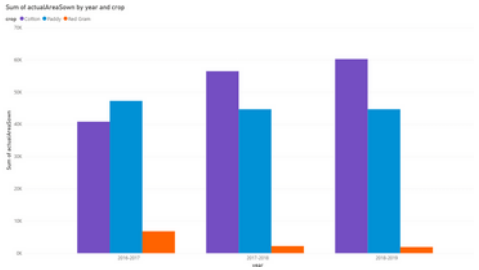
Trend Analysis: Dominance of Cotton and Decline of Paddy Share

Cotton Dominance: Area sown was consistently dominated by Cotton, accounting for up to 60.5% of the total area in 2018-19, cementing its status as the primary cash crop.

Paddy Decline: The proportional share of paddy area declined steadily (from 40.5% to 28.7%), suggesting an ongoing shift of land to cash crops.

The Red Gram Swing: Red Gram area saw a massive spike in 2017 (19.4% of total area) followed by a sharp collapse in 2018-19 (1.0%), indicating a highly responsive, short-term Cropping Pattern Shift.

Fig 5: Cropping pattern shifts of the major crops from 2016-19



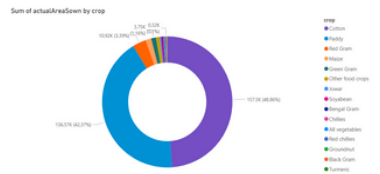

Correlation & Volatility: Identifying Risk Buffers

- Stability Index: Paddy exhibited the lowest area volatility (CV of only 3.3%), confirming its role as the stable foundation crop.
- High-Risk Swing Crop: Red Gram showed extreme volatility (CV of 75.1%), highlighting its status as the region's most flexible crop, which is highly sensitive to annual market and weather conditions.
- Area Trade-off: The inverse correlation in area share between Cotton and Paddy suggests a direct competition for land resources, especially in favorable years.

Table 1: CV analysis

| Crop | Average Area Sown(Ha) | Std. Dev. (Ha) | Coefficient of Variation (CV) | Volatility Interpretation |
|---|---|---|---|---|
| Red Gram | 3638.9 | 2731.3 | 75.1 | Extreme |
| Maize | 1507.9 | 511.3 | 33.9 | Moderate-High |
| Cotton | 52498.3 | 10306.4 | 19.6 | Moderate |
| Paddy | 45523.6 | 1509.8 | 3.3 | Low |

Fig 6: Pie chart showing the biggest crop shares

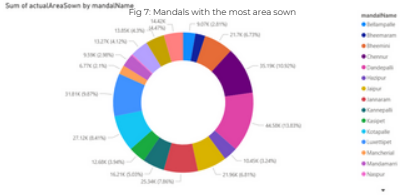



## Conclusion

Defining the Farming Strategy

- Mancherial's agricultural strategy is characterized by a stable base of Paddy (for food security) overlaid by a high-value, dominant Cotton economy.
- The volatile pulses (Red Gram) serve as the primary short-term flexibility buffer, allowing farmers to adjust quickly to price signals or policy incentives.
- The long-term trend shows a clear prioritization of high-value crops (Cotton) at the expense of staple/food crops (Paddy).

Fig 7: Mandals with the most area sown




## Recommendations

Policy Implications and Forecasting

- For Stability: Implement robust water and price assurance policies for Paddy to halt the diversion of land and safeguard regional food security.
- For Pulses (Flexibility): Leverage the high flexibility of Red Gram by announcing attractive Minimum Support Prices (MSP) well in advance of the Kharif season to ensure adequate national pulse production.
- Forecasting: Future area allocation is strongly tied to global cotton and domestic textile market forecasts; these should be the primary inputs for agricultural planning.
- Diversification: New incentive schemes are needed to make mid-volatility crops like Maize economically viable against the powerful pull of Cotton.

## Acknowledgements

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